

COLOSSEUM · SOLANA

Vitals

A patient is dying on a clock and you decide what happens next.

Anyone can play. Nobody can fake the replay.

EP1 · THE LAST BITE

THE PRODUCT

**A nineteen-year-old is dying on a clock.
You are the doctor. Every run leaves a proof.**

Vitals EP1 · THE LAST BITE · F9DBB7DE05 [← episodes](#) [restart](#) [resume](#) [easy](#) LOCALNET · TREE #40348 · 3 ANCHORED · 0 YOURS



MONITOR Deteriorating

HR
128

BP
83/54

SpO₂
94%

RR
28

Monitor

Vent

Pump

BED 3 · ER Ing · F 19 · Deteriorating 2 ALARM 14:21:30

SINUS 25 mm/s

PLETH

RESP

HR 128 bpm

SpO₂ 94 %

NIBP 88/54 mmHg · 16 s ago

RR 28 imp.

TALK TO HER

She answers in her own words, from a model running on our own hardware. She can be wrong, and she can be too breathless to answer.

TREAT HER

Orders, drugs, airway, fluids. A real bedside monitor moves the way the physiology moves.

SHE LIVES OR SHE DOES NOT

No script. A deterministic clinical automaton decides, in real time, from what you did and when.

THE RUN IS PROVABLE

Every case reduces to one hash. Anyone can replay the tape and get the same answer, without asking us.

Eighteen of Thailand's twenty-nine medical faculties already run the engine underneath this.

Vitals is the verifiable layer on **Embla** — a clinical simulation engine live for three months, adopted without a sales team, and awarded 1st Place at NECTEC AI for Thai 2026. Live right now, in production: embla.megawiz.co.th



18 / 29

Medical faculties with active learners — counting verified faculties, not self-reported signups

290+

Clinicians and students — individual accounts, not institutional seats

671

Scored runs across 433 authored clinical scenarios — as of 25 Aug 2026

1st

Place at NECTEC AI for Thai 2026 — announced by NECTEC

THE TEAM THAT SHIPPED IT

PARIPOL TOOPIROH

CEO & AI architect — Embla core engine, Rust end to end

VORANITCHAYANUN
PANTHATACHAPHISARN (OUM)

Co-founder & COO — registered nurse; hospital ops & SOP co-design

PONGSAKORN JAIKAEW

AI engineer — persona engine, ASR/TTS, automated evals

NAPAPATCH VIPUTHBOONYAPAT

BA & QA lead — scenario quality, E2E, on-site testing

Clinical advisor — Dr Preechayuth Yokanit — anaesthesiologist. MD, Phramongkutklao College of Medicine, Mahidol University (2000); Diploma of the Thai Board of Anesthesiology, King Chulalongkorn Memorial Hospital (2004). Cases pass his review before release.

Reviewer — Pimnipa Chaosirikul — learner-experience reviewer; final-year medical student, Faculty of Medicine Siriraj Hospital, Mahidol University. Reviews realism and wording, not clinical correctness.

The adoption is Embla's, earned before any of this was anchored. What Vitals adds is the part a school cannot do for

THE PROBLEM, AND THE MISSION

One doctor for every 1,487 people in Thailand. One for every 1,650 in Indonesia.

Our mission: one per cent more doctors who actually reach patients, every year — *licensed* doctors, not merely graduates.

PEOPLE PER DOCTOR · PHYSICIANS WORKING IN THE HEALTH SYSTEM PER 1,000 POPULATION, 2024 · NATIONAL SOURCES: MHLW (JAPAN), MOPH (THAILAND), KEMENKES (INDONESIA) · COUNTS OF DOCTORS DELIVERING CARE, NOT OF LICENCE HOLDERS; EACH COUNTRY'S OWN POPULATION BASE

1:370

Japan · 2.7 per 1,000 · MHLW, 2024

1:1,487

Thailand · 0.67 per 1,000 · MOPH, 2024

1:1,650

Indonesia · 0.61 per 1,000 · Kemenkes, 2024

A Thai doctor carries four times the load a Japanese one does — and an Indonesian doctor carries that same load across four times as many people. The problem does not get easier next door; it gets bigger. **A product that charges aims itself at the places it matters least.**

One caveat we would rather say than have found: Thailand's denominator is registration-based and leaves out several million non-registered residents — the UN puts Thailand at 71.7 m against MOPH's 65.0 m. On a UN-comparable denominator Thailand is **1:1,639**, not 1:1,487. The figure printed above is the one that flatters Thailand.

WHY 1% IS A TARGET AND NOT A SLOGAN

Attrition is **9.1%** globally — the review that reports it gives a range of **2.7–20.1%**, and a second review puts it at 11.1%. Moving 9.1% to 8.2% yields 1% more graduates, so **1% means cutting attrition by about a tenth** — 1 in 10 of everyone who leaves today, whatever their reason. The reason cited most often is **academic difficulty, at 55.7%**, and it is the one we act on — cited, not apportioned: the four reasons in that review overlap and sum to 145.7%, so no share of dropouts can be read off it. Order-of-magnitude effects already exist here: in one South American cohort (Vergel et al.), curriculum reform cut dropout from **41.5% to 3.3%**. We need a tenth of that.

A number we used to lead with, kept with the wording it needs: across 3,142 US counties, every 10 additional *primary care* physicians per 100,000 people was *associated* with 51.5 more days of life expectancy — roughly 1,400 person-years across that whole population (Basu et al., *JAMA Internal Medicine* 2019). It is a US primary-care association measured over populations, not an effect any one doctor produces.

GAMEPLAY & VERIFIABLE OUTPUT

The replay is the trophy, not the score.



GHOST · MOCKUP

you · adrenaline 1:12 ghost · 0:38

Ghost racing. The tape replays exactly, so another player's run can run beside yours — and you watch the two patients diverge while you are still deciding. Trackmania's ghost car, in an emergency room.



RECORD · MOCKUP

harm · stand/walk collapse 50 saves · 12 harms

Harm is on the record. Fifty saves with twelve harms reads differently from thirty with none — and it keeps the fastest run and the cleanest run from being the same run.



DEATH CARD · MOCKUP

DeathArrest · turned at 2:14

gold path → adrenaline IM at 0:40

Failure is content. "She died because you reached for antihistamines" is a better clip than any win. A loss should be legible, teach in the same breath, and be worth showing someone.

treated adrenaline IM · O ₂ · supine · fluids · admit	WinDischarge · leaf: 2bb3ce...
stood up adrenaline IM · stood the patient up · O ₂ · fluids · admit	WinDischarge · harm · leaf: e6ef0c...
no adrenaline antihistamine · steroid · wait	DeathArrest · fatal · leaf: 4be24a...

Deterministic execution: Three players, three tapes, replayed against the shipped physiology automaton — cargo run -p vitals-replay prints these three. Same outcome as the first, different leaf — because harm is on the record.

ONE SEASON · EP1 IS FILMED, THE REST ARE AUTHORED AUTOMATA

- EP1

The Last Bite

anaphylaxis · she has minutes
- EP2

Time Is Muscle

STEMI · the clock is muscle
- EP3

Don't Make Him Cry

epiglottitis · doing nothing is the treatment
- EP4

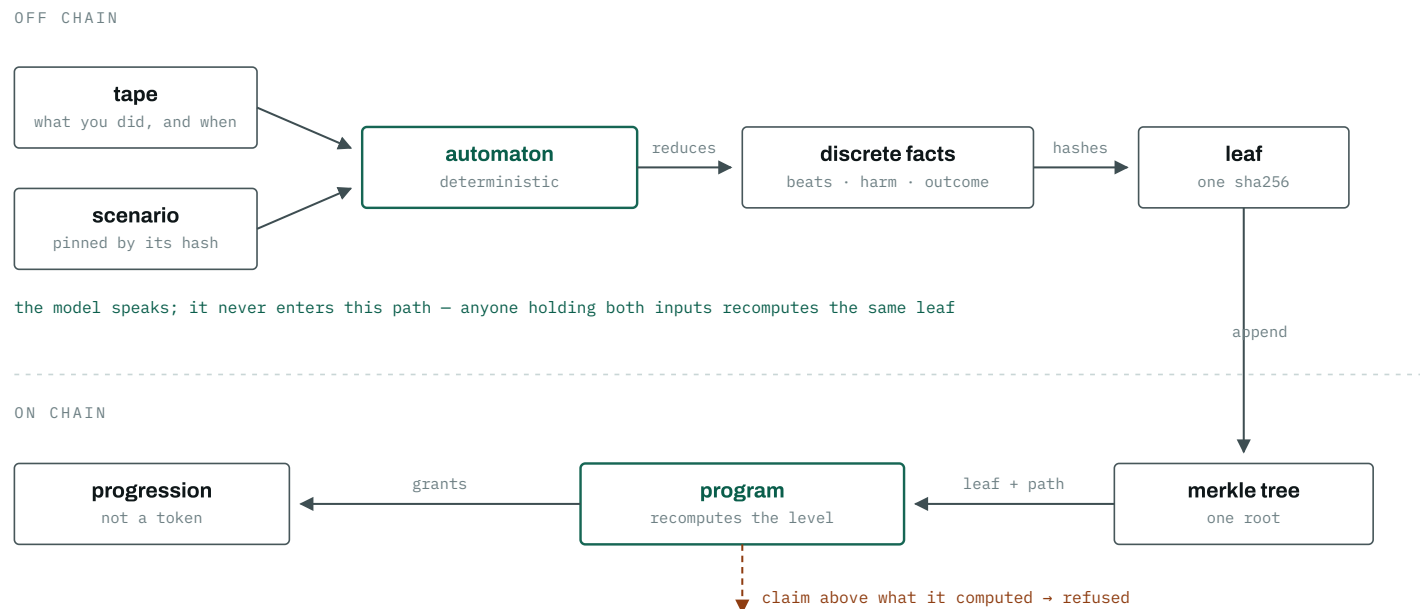
The Masquerader

PE · everything looks normal
- EP5

The Night the Stars Fell

mass casualty · the queue is the cost

The model makes it playable. The automaton makes it provable. Two inputs in, one hash out — and the program refuses its own user.



The tape and the scenario are the only inputs. The chain never sees a run — only a leaf that has to prove against a root it built itself.

LIVE VALIDATOR · CLAIM: PROFICIENT

REJECTED — claimed Proficient, computed Competent (3 distinct attempts)

LIVE VALIDATOR · CLAIM: COMPETENT

GRANTED — stored Competent, xp 35. Stored level is what program computed.

No issuer key. No oracle. The Solana program recomputes the level from Merkle proofs and writes only what its own arithmetic agrees with.

Free to the learner. Paid by whoever relies on the proof.

The learner never pays — we tested charging, **eighteen faculties use it and none bought**, and charging would aim this at the countries that need it least. The economics live on the other side: **a doctor’s proof that travels** — checkable in any country, by anyone, without asking any institution — and **every act of reliance is a paid, on-chain event**.

THE LOOP — FOUR ACTORS, ONE RAIL

learner	plays, practises, anchors a record	— free —
relying party	residency programme verifies a candidate	pays USDC
case author	publishes a station anyone can train ...	paid per proven replay
verifier	would replay the tape — bonded, slashed if it lies	paid the fee

More learners → more records worth relying on → more paid verification → more authors paid → more cases → more learners. **The flywheel runs on reliance, not speculation.** Today the learner row runs and the author row is built — on devnet, off on the public bay until a rate is set, and a plain wallet transfer rather than a program instruction: the program holds no money and moves none. The relying-party and verifier rows are **designed, not built**.

WHAT ONLY A CHAIN DOES HERE

BORDERLESS BY CONSTRUCTION

The relying party is in another country. A shared neutral layer is the only database both sides can trust — take the chain away and you are back to one company’s word.

DONATIONS THE PROGRAM COULD ENFORCE

Coming next: money given for a scenario could only be spent on that scenario, released against the delivered content hash, so a funder recomputes our impact instead of reading our report. The program holds no money today; that instruction is not written yet.

AUTHOR PAY THAT CROSSES BORDERS

Built on devnet, not yet on the public bay: case authors are paid per proven replay in lamports, settled in seconds — amounts a card rail’s fixed fee would swallow whole. This is what makes an open case marketplace possible at all; the Case Registry that would put authorship itself on-chain is still a drawing rather than a shipped instruction.

ON THE TOKEN QUESTION, BOTH LAYERS IN ONE BREATH

There is no token. Progression and credentials are **deliberately untradeable** — progression is a record the program owns rather than a mint, so there is nothing to sell, and a market in “Expert in Cardiology” would void the product. That half is built. The other half is a drawing: **\$VIGIL** *would* be a separate verification-market token that **players never hold and never see**, bonding the operators who replay tapes and slashed when one lies, with the arena named **3R** so the two never share a name. **Designed, not built** — no mint, no bond account and no slashing instruction exists in this repository, and any fungible token is an explicit scope cut for this sprint.

This shape already won this hackathon once.

FRONTIER 2026 · GRAND CHAMPION

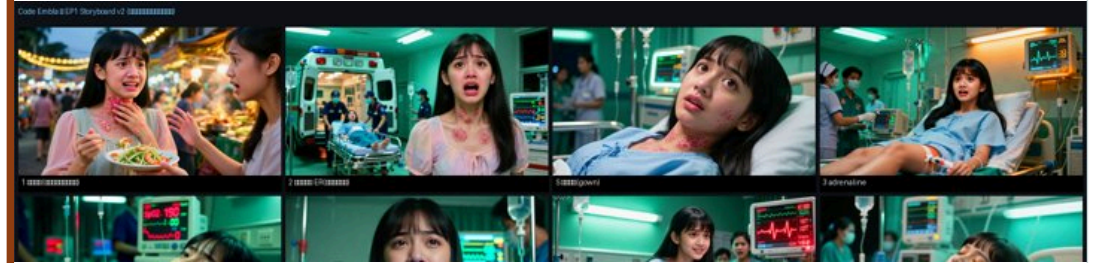
CrowdBrain — as Colosseum described them: *"train users in simulation, qualify them through QA, and route the best operators to real robots."*

The exact same shape — proven at the largest crypto hackathon ever run (2,857 submissions, 150 countries) for robot operators instead of clinicians.

PATTERN: VERIFIABLE HUMAN IN THE LOOP

HOW EPISODE SIX GETS MADE — SKALD CONSOLE

AI SHOWRUNNER + QA



Skald verifies the content. Solana verifies the play.

Both projects sell the same thing: human work made checkable before anyone relies on it. Their own words to robotics teams: *"Qualified manipulation operators, approved robot nodes, and verified data — without building it in-house."* That is verification by approval. Ours is **verification by arithmetic**: replay is exact, so two honest verifiers cannot disagree and a lie is caught by recomputation. The bond that would pay a verifier is ours alone, and **designed, not built**.

WHY MEDICINE FIRST

Highest stakes & regulation. Solve trust here and it generalises downward.

WHY SOLANA

Today's demo: 30,000 lamports a run — 30 SOL per million, measured on devnet · the production design is compressed anchoring at 5,729–10,729 lamports (relay alone, or the learner co-signing as today), computed on a local validator and not yet on devnet · authors are paid per proven replay, settled in seconds — built on devnet, off on the public bay until a rate is set · the learner is gasless and never sees a seed phrase. The numbers only close on this chain.

HEAD START (3 MOS LIVE)

18 of 29 Thai medical faculties · 290+ clinicians and students · 1st Place @ NECTEC 2026.

WHERE IT STANDS, AND WHAT'S NEXT

Two halves built. One sprint, 29 Aug – 26 Sep, joins them — then the arena.

WORKING NOW (LIVE IN PRODUCTION)

- **Replay** → **leaf**. EP1 runs against shipped automaton; three tapes, three distinct leaves, reproducible across runs.
- **Program** → **refusal**. Native Solana program **live on public devnet** (535FMHHZ...BTSG), recomputing levels and rejecting invalid claims — deployed bytecode verified byte-for-byte against the build.
- **Award-Winning Engine**. 1st Place @ NECTEC AI for Thai 2026, announced by NECTEC (facebook.com/NECTEC/embla.megawiz.co.th).
- **Rust end to end**. One scoring implementation compiled into game, verifier, and onchain program.

Cut on purpose

STATED BEFORE ANYONE ASKS

40 of 100 points are re-derivable; 60 are attested — signers name the model and version, and more signers means more confidence, never proof. **We cannot prove the human at the keyboard was the enrolled student**; identity binding is delegated to the issuer, and the credential carries the issuer's name so a relying party can price that issuer's rigour. **We do not grant standing** — only a board can make competence authoritative. And **the pass-rate number does not exist yet**: what exists is the measurement reliability, the study design, and the date it lands.

ZK selective disclosure · verifier DePIN with live staking · token launch · mobile · new clinical content. Each is a real v2 line and none survives the run-up. The credential layer stays a stub until an accrediting institution is standing next to it — progression doesn't need one, which is why progression is the demo.

THE SPRINT BEFORE THE ARENA OPENS

COMMIT-REVEAL, IN CODE

The record gains the commitment and the slot it was made at, so the proof travels in the leaf rather than depending on which program version ran.

THE OSCE SCORER, PORTED CLEAN

The deterministic 40 as a dependency-free crate, bound to the reference engine by its own conformance vectors. The judged 60 stays outside the auditable core, where it belongs.

THE UNLOCK THE CHAIN CHECKS

Stars earned → the program recomputes the predicate → the next scenario opens. Legible without knowing any medicine.

LIVE, PUBLIC, AND METERED

vitals.academy open to strangers with the first scenario free and no account — because four weeks of real usage cannot be written the week before a deadline.

The 60-second version: you play a patient who is dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.

The ask: the people who would actually rely on a record like this. We can build the protocol — we cannot grant it standing.